

Field and Galois Theory

(Back Paper, BMath-3rd year, 2025)

Instructions: Total time 2 Hour. All problems are compulsory. Use terminologies, notations and results as covered in the course, no need to prove such results. If you wish to use a problem given in a homework or an exercise from the class, please supply its full solution.

1. Let p be a prime and x, y be indeterminates (variables) over $\mathbb{F}_p$, $E = \mathbb{F}_p(x, y)$, $F = \mathbb{F}_p(x^p, y^p)$. Show that **(i)** $[E : F] = p^2$. **(ii)** E has no primitive element over F .
2. Let p be an odd prime and $a \in \mathbb{Q}$ be such that a is not a p th power in $\mathbb{Q}$. Let $f(X) = X^p - a$. Prove that **(i)** $f(X)$ is irreducible over $\mathbb{Q}$. **(ii)** Compute the Galois group of $f(X)$ over $\mathbb{Q}$.
3. Let $f(X) \in \mathbb{Z}[X]$ be a degree 4 irreducible polynomial with Galois group S_4 over $\mathbb{Q}$. Let $\alpha \in \mathbb{C}$ be a root of f . Prove that $\mathbb{Q}(\alpha)/\mathbb{Q}$ has no subextensions of degree 2.
4. Let $f(X) \in \mathbb{Q}[X]$ be a polynomial of degree n having distinct roots $\alpha_1, \dots, \alpha_n$ in a splitting field. Let $\delta(f)$ denote the discriminant of f .
(i) Prove that $\delta(f) = (-1)^{n(n-1)/2} \prod_{i=1}^n f'(\alpha_i)$, where f' denotes the formal derivative of f .
(ii) Let p be an odd prime and $\lambda_p(X) = 1 + X + \dots + X^{p-1}$. Prove that $\delta(\lambda_p(X)) = (-1)^{\frac{p-1}{2}} p^{p-2}$.